

Reporting, Reproducibility and the Results Section

150 minutes · 6 self-check questions

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Turning output into a results section a reviewer will accept: what to report, how to lay out tables, and how to make the whole analysis reproducible by someone else.

Learning objectives

- Write a results section that matches the pre-specified analysis plan
 - Report estimates, intervals and p-values to the right precision
 - Lay out tables and figures to journal standards
 - Apply the relevant reporting checklist
 - Build a reproducible analysis with a script and a clear folder structure
 - Produce a dynamic report with R Markdown or Quarto
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What belongs in the results

- Participant flow: how many were approached, included, excluded and analysed, and why
 - Table 1 of participant characteristics
 - The primary analysis, answering the primary objective, first
 - Secondary analyses clearly labelled as secondary
 - Any analysis not planned in advance labelled exploratory — not doing so is the difference between honest reporting and fishing
 - No interpretation in the results section; that belongs in the discussion
 - Every objective from the proposal produces a result, and every result answers an objective
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Precision and notation

- Report to the precision the data support: a mean age of 42.3 years, not 42.3187
- Percentages to at most one decimal, and never for a denominator under about 20
- p-values to two or three decimals; below 0.001 write $p < 0.001$, never $p = 0.000$
- Effect estimates with confidence intervals, using "to" rather than a dash when values may be negative
- Give the unit of measurement in the row or column label, not repeated in every cell

- Be consistent about decimal places within a column so figures line up

COMMON REPORTING FAULTS

Written	Problem	Better
$p = 0.000$	No p-value is zero	$p < 0.001$
Mean 42.3187 years	False precision	Mean 42.3 years
OR 2.4 ($p < 0.05$)	No interval	OR 2.4 (95% CI 1.3 to 4.5)
33.3% ($n = 3$)	Percentage of a tiny denominator	1 of 3
A trend towards significance	$p = 0.08$ is not a trend	No significant difference ($p = 0.08$)

Checklists

- STROBE for observational studies — cohort, case-control and cross-sectional
- CONSORT for randomised trials
- PRISMA for systematic reviews
- TRIPOD for prediction model studies
- SAMPL for statistical reporting generally
- Find them at the EQUATOR Network, and use one while writing rather than afterwards
- Most journals require the completed checklist at submission anyway

Making the analysis reproducible

- A reproducible analysis means someone else can run your script on your data and get your numbers
- Structure: data/raw (read-only), data/clean, scripts, output, and a README explaining the order
- Number scripts in running order: 01-clean.R, 02-describe.R, 03-model.R
- Never edit data by hand; every change is a line of code
- Set a seed before anything random so imputation and bootstrap results repeat exactly
- Record package versions — `sessionInfo()` in R, or `renv` for a full lockfile
- Use relative paths so the project runs on another machine

IN PRACTICE

The practical test: delete every output file, rerun the scripts from raw data, and check the numbers in your manuscript still match. Most analyses fail this the first time, which is exactly why it is worth doing before a reviewer asks.

Dynamic reports

- R Markdown and Quarto embed the code that produces each number in the document itself
- The manuscript is then generated from the data, so a table can never disagree with the analysis
- Inline code writes results into prose: the mean was ``r round(mean(dat$age), 1)`` years
- Re-running after a data correction updates every number, table and figure at once
- Output to Word for co-authors, PDF for submission, or HTML for sharing
- This removes the single most common source of error in a results section: copying a number by hand and later changing the analysis

Worked examples

A results paragraph rewritten

A draft reads: "The intervention group had significantly lower blood pressure ($p < 0.05$), showing that the programme works well and should be rolled out nationally."

1. No estimate: how much lower? The reader cannot tell if it is 1 mmHg or 20.
2. No interval: no sense of precision.
3. $p < 0.05$ rather than the exact value, which discards information.
4. "Showing that the programme works well" is interpretation, and belongs in the discussion.
5. "Should be rolled out nationally" is a recommendation, and belongs in the conclusion — supported by more than one trial.
6. Rewritten: "Mean systolic BP was 8.4 mmHg lower in the intervention group (95% CI 3.1 to 13.7; $t(118) = 3.12$, $p = 0.002$; $d = 0.57$)."

THE POINT

The results section reports what was found, in numbers, with uncertainty. Every clause of interpretation that creeps in is a clause a reviewer will ask you to move.

The table that no longer matched the analysis

Two weeks before submission a data error is found and corrected. The analysis is rerun. The manuscript is submitted, and a co-author notices Table 3 still carries the old numbers.

1. The tables had been typed by hand from earlier output.
2. Rerunning the analysis updated the output files but not the manuscript.
3. Three of eleven numbers in Table 3 changed; two did not, so nothing looked obviously wrong.
4. In Quarto or R Markdown the table is generated from the data at render time, so this cannot happen.
5. Inline code puts results into prose too: the mean was ``r round(mean(dat$age), 1)`` years.
6. Re-rendering after the correction would have updated every number, table and figure at once.

THE POINT

Copying numbers by hand and later changing the analysis is the most common source of error in a results section. Generating the document from the data makes the failure impossible rather than unlikely.

Key terms

Primary analysis

The pre-specified analysis answering the study's main objective.

Exploratory analysis

An analysis not planned in advance, which must be labelled as such.

STROBE

The reporting checklist for observational studies.

Reproducibility

The property that another person can obtain your results from your data and code.

Seed

A starting value for the random number generator, making stochastic procedures repeatable.

Dynamic report

A document in which results are generated from the data at render time rather than pasted in.

Activity

Take one table from a draft of your own results. Check it against the reporting faults table in this unit, fix every instance, and add a confidence interval to every estimate that lacks one. Then write the accompanying paragraph without a single word of interpretation.

Further reading

- **EQUATOR Network — reporting guidelines** — Free index of every checklist. Find yours before you write.
- **Lang & Altman — SAMPL guidelines for statistical reporting** — Concrete rules for numbers, tables and notation.
- **Quarto documentation** — Works with R, Python and Stata; the successor to R Markdown.
- **The Turing Way — Guide for Reproducible Research** — Free, community-written, practical on project structure.

Self-check questions

1. Your software reports $p = 0.000$. You should write:
 - a. $p = 0.000$
 - b. $p < 0.001$
 - c. $p = 0$
 - d. p is highly significant
2. You ran an unplanned subgroup analysis that produced an interesting finding. In the paper it must be:
 - a. Reported as a primary result
 - b. Labelled exploratory, with the reader told it was not pre-specified
 - c. Omitted entirely
 - d. Reported without a p-value
3. "There was a trend towards significance ($p = 0.08$)" is poor practice because:
 - a. $p = 0.08$ is not reportable
 - b. A p-value is not a trend; the result either meets the threshold or does not, and the interval is what conveys the uncertainty
 - c. Trends require at least three time points
 - d. The threshold should have been 0.10
4. The practical test of a reproducible analysis is that:
 - a. The code is commented
 - b. Deleting all outputs and rerunning the scripts from raw data reproduces every number in the manuscript
 - c. It was run in R rather than SPSS
 - d. The data are publicly available

5. Setting a random seed before multiple imputation matters because:
- a. It makes the imputation more accurate
 - b. It makes the results exactly repeatable on a rerun
 - c. It is required by Rubin's rules
 - d. It reduces the number of imputations needed
6. The main advantage of writing results in Quarto or R Markdown is that:
- a. It produces prettier documents
 - b. Numbers are generated from the data at render time, so a table can never disagree with the analysis
 - c. It runs the analysis faster
 - d. It removes the need for a statistician
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Answers

1. **b.** The value has been rounded, not computed as zero. A probability of exactly zero would mean the data are impossible under the null, which is never the case.
 2. **b.** Unplanned analyses are legitimate and often valuable, but presenting one as though it had been planned misrepresents the false positive risk. Label it and let the reader weigh it accordingly.
 3. **b.** The phrase implies the effect would have been significant with more data, which is not something the p-value says. Report the estimate and interval and let them speak.
 4. **b.** Reproducibility is demonstrated, not asserted. Rerunning from raw data is the only check that catches hand-edited cells, undocumented steps and numbers that drifted after the analysis changed.
 5. **b.** Imputation is stochastic, so results differ slightly each run. A seed fixes the random sequence so you and a reviewer obtain identical numbers. It does not affect validity.
 6. **b.** Copying numbers by hand and then changing the analysis is the commonest source of inconsistency in a results section. Generating them from the data at render time makes that failure impossible.
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